

Bucket 5

Positive-Carry Tail Insurance

How short-vol carry funds SPX put protection,
why monetizing those puts is the edge,
and what can go wrong.

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Contents

1. The idea in plain English
2. Two layers: live sleeve vs insurance product
3. How the carry engine makes money
4. Why we buy SPX puts with that carry
5. Monetizing the puts — the real edge
6. How it has performed
7. Risks — what can go wrong
8. The tiny live sleeve in the book
9. What actually matters (simplified knobs)
10. Operator takeaways

1. The idea in plain English

Most short-volatility strategies make money slowly in calm markets and then give it all back — or worse — in a crash. Bucket 5 flips that script.

The insurance design is meant to behave like a **self-funding insurance policy**:

- In quiet markets, shorting volatility ETPs (mainly UVIX, hedged with SVIX) earns a steady carry from futures roll and the volatility risk premium.
- Only a fraction of capital sits in that short-vol book. The rest sits in T-bills, earning a safe yield.
- Part of the combined income buys a ladder of longer-dated SPX puts.
- When equities crash and vol spikes, those puts surge. The strategy **sells into that surge** — monetizes — so the crash becomes a cash payout instead of a paper gain that later evaporates.
- Harvested cash is partly redeployed into short-vol again when the term structure is stressed (when forward carry is richest) and partly parked back in bills.

*The goal is not maximum short-vol return. The goal is a book that **pays you to wait** and then **pays you again when the world breaks**.*

2. Two layers: live sleeve vs insurance product

"Bucket 5" in this repo covers two related things. They share the UVIX/SVIX complex and the same worldview, but they are not the same trade.

Layer	What it does today	Scale
Live sleeve	A tiny carve-out inside the daily L/S ETF book. Short a vol ETP (e.g. UVIX), hedge with the pair underlying. Sized like a small inverse pair — not dual-short, not put-funded.	~0.25% of book gross (~\$10.5k on a \$4.2M book)
Insurance product	Research / lab design: short UVIX and short SVIX, T-bill collateral, adaptive rebalancing, SPX put ladder, monetize & redeploy. This is where the insurance thesis lives.	Standalone backtest capital (lab / dashboard)

Most of this guide is about the insurance product — because that is where the SPX monetization story, the performance profile, and the real risk discussion live. The live sleeve is covered briefly at the end.

3. How the carry engine makes money

UVIX is a leveraged long-vol product. In a normal contango VIX futures curve, it bleeds value over time as contracts roll. Shorting UVIX harvests that bleed. The problem: in a vol spike, UVIX can explode upward, and a naked short can blow through any amount of prior carry.

SVIX is roughly the opposite exposure (short-vol). Shorting SVIX loses money in calm contango but **pays when vol spikes**. So the insurance book shorts **both**, and uses a simple mix dial:

- **More short UVIX than SVIX** when the curve is steeply in contango (calm). That maximizes carry.
- **More short SVIX (toward a vol-neutral mix)** when the curve flattens or goes into backwardation (stress). That reduces the linear blow-up risk and cuts overall size.
- Only about **20% of equity** is deployed in this dual-short sleeve. The other ~80% earns T-bill yield. That keeps the short-vol book from dominating the whole account.

*Intuition, not formula: think of the dual-short as a **carry engine with a built-in shock absorber**. It still has residual crash risk — that is intentional. The residual tail is what the SPX puts are paid to cover.*

Rebalancing also speeds up in stress. In calm markets the book may sit for roughly two weeks between adjustments. As the VIX term structure worsens, it rebalances much faster so the mix and size can move into the spike instead of drifting on an old calm-market setting.

4. Why we buy SPX puts with that carry

Carry alone is not insurance. Carry alone is a short-vol bet that hopes crashes are rare. The insurance product deliberately spends part of its income on a **ladder of longer-dated SPX puts** — several strikes, bought around six months out and rolled around three months.

Why a ladder instead of one put?

- A near put (e.g. ~10% out of the money) responds earlier in a selloff.
- Deeper puts (20–30% out of the money) are cheaper and pay more in a true crash / volmageddon-style move.
- Spreading premium across rungs avoids betting the whole hedge budget on one strike being "the" crash level.

In production settings the book spends on the order of a couple percent of equity in put premium per roll cycle — funded by sleeve carry plus T-bill income, not by selling the farm. When puts are cheap (deep contango, calm VIX), the design spends a bit more. When the curve is already stressed, it throttles new premium because existing puts are already live.

*Owning puts is only half the story. **Holding puts through the entire spike and then watching them decay as markets bounce is how most "hedged" short-vol books still lose the plot.** Monetization is what converts a temporary mark-to-market windfall into permanent capital.*

5. Monetizing the puts — the real edge

This is the section that matters most. The SPX ladder is not a set-and-forget hedge. It is a **harvesting machine**.

5.1 What "monetize" means here

When equities fall and implied volatility jumps, put prices can multiply. A put bought for \$1 might trade at \$3, \$5, or \$8 of mark-to-market value. If you do nothing, that value often collapses on the bounce — vol mean-reverts, spot recovers, and the "hedge that saved you" quietly gives the money back.

Monetization means: **sell a planned fraction of the put position into strength**, lock cash, and optionally re-arm a fresh cheaper put at the new lower spot. The crash stops being a temporary accounting gain and becomes a banked insurance payout.

5.2 How the harvest rules work (conceptually)

The rules are deliberately simple and mechanical — no discretionary "feels expensive" calls:

- **Profit tiers.** As a put's value crosses multiples of its cost (roughly 3×, then 5×, then 8×), sell a chunk of what remains. Early tiers take partial profits; the highest tier can clear the position.
- **VIX override.** If VIX itself hits extreme levels (around 45, then 65), monetize even if the profit-multiple path was uneven. Outright panic is its own signal.
- **Giveback stop.** Once a put has already doubled, if it then falls meaningfully from its peak (about a third), sell down. This stops the classic failure mode: a 6× winner that round-trips back to 1× while you "wait for more."
- **Re-arm.** On a full exit, bank most of the proceeds and optionally buy a fresh longer-dated put at the new spot — so a second leg down can still pay.

5.3 Why monetization changes the return shape

Without monetization, puts are a drag in calm years and a temporary cushion in crashes. With monetization, calm years still pay for the premium (carry + bills fund it), but crash years can produce **realized cash** that compounds afterward.

That cash is not left idle by default. After a harvest — which usually coincides with a stressed vol curve — the design leans harvested money back into the short-vol sleeve (when forward carry is richest as vol mean-reverts) and parks the rest in T-bills. In calm regimes it keeps more dry powder in bills. The economic story is: **get paid by the crash, then get paid again by the reversion.**

In the live-era production backtest (Mar 2022 – Jul 2026, \$1M start), monetization events fired over 100 times across profit tiers, VIX triggers, givebacks, rolls, and re-arms. Cumulative realized put proceeds were on the order of ~\$840k of harvested cash, with additional growth from redeploying that cash — i.e. the puts were not just a deadweight hedge cost.

6. How it has performed

Figures below are from the dashboard panel for the production insurance preset (dynamic 2.4% put budget), live UVIX/SVIX era, ThetaData put pricing where cached. They are research results — not a promise for the tiny live book sleeve.

Lens	What happened (live era ~2022–2026)
Carry alone	The dual-short sleeve by itself earned only a few percent per year. That is intentional: size is capped (~20% of equity), and the mix de-risks in stress. Carry is the funding source, not the hero.
Combined book	Sleeve + T-bills + put ladder + monetize/redeploy produced roughly ~14–15% CAGR, Sharpe around ~1.0, and max drawdown around ~6–7% in this sample.
August 2024 vol spike	The combined book was up ~21% over that month — a period when naked short-vol strategies typically suffer. Monetized puts are the main reason the sign flips.
Crash stress tests	Instantaneous scenario payoffs (carry P&L + put value) were strongly positive: roughly +33% in a mild –20% SPX shock, +90% in a –30% shock, and ~+175% in a volmageddon-style –40% shock (research scenarios, not live fills).

Read the performance the right way: the strategy is not "beating the S&P." It is trying to deliver **positive carry in peacetime and positive payout in wartime**, with drawdowns that look more like an insurance book than a levered short-vol fund.

Also keep scale straight. These insurance results assume a dedicated capital base running the full product. The live GTP sleeve is only ~0.25% of the multi-strategy book and does not currently run the put ladder in production execution.

7. Risks — what can go wrong

A strategy that monetizes crash convexity still has serious failure modes. These are the ones that matter.

7.1 Path and gap risk on the short-vol legs

UVIX can gap violently overnight or through a liquidity vacuum. Even with SVIX as a partial shock absorber and with size cut in backwardation, the dual-short sleeve can lose several times its calm-day P&L in hours. Puts help on a multi-day equity crash; they do not perfectly offset a one-print VIX ETP dislocation or an ETF trading halt.

7.2 Monetization can be early — or too late

Selling at 3× feels smart until the put goes to 10× the next week. Keeping everything for the "big one" feels smart until the bounce erases the gain. The tiered rules are a compromise, not a crystal ball. A fast V-shaped crash-and-recovery can monetize little if prices never pause long enough at the tiers, or can giveback-stop you out of a position that later re-spikes.

7.3 Premium drag in long calm regimes

If markets grind higher for years with cheap, decaying puts and only mild vol, the ladder is a recurring cost. Carry plus T-bills are meant to fund that cost, but funding can fail if borrow on UVIX/SVIX rises, slippage worsens, or the term structure stops paying. A long calm stretch is not "risk free" — it is slow bleed risk on the hedge.

7.4 Correlation and basis risk

The hedge is SPX puts. The carry engine is VIX-futures ETPs. Those usually spike together in equity crashes, but not always one-for-one. A vol-of-vol event, a VIX ETP structural change, or a crash that is more single-name / credit than index can leave the puts under-helping the sleeve loss — or vice versa.

7.5 Liquidity, borrow, and product risk

UVIX and SVIX can be hard to borrow, expensive to short, or altered by issuer rules in stress. Options markets gap; bid/ask on deep puts widens exactly when you want to monetize. Backtests that use mid prices will overstate

harvest quality. Issuer closures, reverse splits, and tracking-error regime shifts are real operational risks.

7.6 Model and sample risk

Live-era history since UVIX/SVIX inception is short. Extended samples lean on synthetic history. Crash scenario payoffs are stylized. Parameter choices (how fast to de-risk, how much premium to spend, when to take profits) were tuned on the same world they are measured in. Out-of-sample humility is required.

7.7 Redeploy risk after a harvest

Leaning harvested cash back into short-vol after a spike is the "get paid by reversion" trade. If vol keeps rising — a second leg down — that redeploy can give back part of the insurance win. That is why the design keeps a large bill allocation in calm times and why runner-put experiments exist: to leave some convexity on for the second hit.

8. The tiny live sleeve in the book

In production, Bucket 5 is a **0.25% of gross** carve-out from the broader inverse/vol family (which totals 3%). It exists because VIX ETPs are not ordinary inverse LETFs: simple decay math misses futures roll and vol risk premium, and they used to be awkwardly haircut inside Bucket 4.

- Universe: VIX-complex names (UVIX, SVIX, UVXY, SVXY, VXX, and related).
- Live structure today: standard inverse-style pair (short the vol ETP, hedge with the underlying leg) — **not** the dual-short + put product.
- Sized with the same decay/edge scoring family as Bucket 4, but without Bucket 4's dynamic hedge engine, ratchet, or crash-budget clamps.
- Borrow and edge gates still apply; expensive or low-edge names go to purgatory (zero new size, not forced close).
- Purpose in the live book: a small, explicit risk budget for vol ETPs instead of burying them inside Bucket 4.

*Think of the live sleeve as the **placeholder and risk budget** for the vol complex. The insurance product is the fuller expression of the thesis — still research until explicitly promoted into live execution.*

9. What actually matters (simplified knobs)

You do not need a parameter encyclopedia to understand the strategy. A handful of choices drive almost all of the behavior:

Knob	Plain-English role
How much capital in the short-vol sleeve	Smaller = less carry, less blow-up. Production research uses about one-fifth of equity; the rest earns bills.
How the UVIX/SVIX mix shifts in stress	Calm → lean into UVIX carry. Stress → shift toward SVIX hedge and cut size. This is the shock absorber.
How much put premium you spend	Too little and crashes still hurt. Too much and calm years bleed. Production aims for a modest percent of equity per roll, spending more when puts are cheap.
How aggressively you monetize	Earlier harvests lock cash but can miss a bigger spike. Later harvests capture more convexity but risk giveback. Tiers + VIX triggers + giveback stops are the compromise.
Where harvested cash goes	More into short-vol after a spike (reversion trade); more into bills when calm. This turns insurance payouts into a second return stream — and a second risk.
Live book size	0.25% of total gross. Deliberately tiny until the insurance stack is promoted beyond the lab.

10. Operator takeaways

- Bucket 5's intellectual core is **fund puts with short-vol carry, then monetize those puts in crashes** so insurance becomes cash, not a temporary mark.
- Performance in the live-era research sample looks like a funded hedge book: mid-teens CAGR, ~1 Sharpe, single-digit max drawdown, and **positive** returns in the Aug 2024 vol event.
- The biggest conceptual risk is believing the puts alone make you safe. Without monetization discipline, put gains often mean-revert away. With monetization, you introduce timing and redeploy risk instead.

- The live GTP sleeve is not yet the full insurance product. Do not read lab crash payoffs as if the \$10k book sleeve will behave the same.
- When reviewing results, ask three questions: Did carry fund the puts? Did monetization bank the spike? Did redeploy respect a second-leg down? Those three answers explain most of the P&L.

Bottom line: Bucket 5 is an attempt to sell patience and buy panic — and to cash the panic before the market takes it back.